

3.3 Kinematics in 2 Dimensions

Mathematics B (MEI) — OCR B — A-Level (H640) — Mechanics — spec refs k9–k11

What this topic covers. Extending kinematics to two dimensions using vectors — position, velocity and acceleration as vector functions of time, the vector forms of the constant acceleration formulae, relative position and velocity, and finding the Cartesian equation of a path.

1. Vector kinematics

In two dimensions each quantity becomes a vector with **i** and **j** components (spec ref k9).

Quantity	Symbol	Relationship
Position	r	Measured from a fixed origin
Velocity	v	$\mathbf{v} = \frac{d\mathbf{r}}{dt}$
Acceleration	a	$\mathbf{a} = \frac{d\mathbf{v}}{dt}$

Going the other way, $\mathbf{v} = \int \mathbf{a} dt$ and $\mathbf{r} = \int \mathbf{v} dt$, each with a vector constant of integration fixed by the initial conditions.

Differentiation and integration act **component by component**, which is what makes two-dimensional motion tractable: the **i** and **j** directions behave independently.

Worked example. A particle has $\mathbf{r} = 3t\mathbf{i} + t^2\mathbf{j}$. Find its velocity, speed at $t = 2$, and acceleration.

$$\mathbf{v} = 3\mathbf{i} + 2t\mathbf{j}$$

At $t = 2$, $\mathbf{v} = 3\mathbf{i} + 4\mathbf{j}$, so the speed is $|\mathbf{v}| = \sqrt{9 + 16} = 5$.

$$\mathbf{a} = 2\mathbf{j}$$

Worked example. A particle has $\mathbf{a} = 2\mathbf{i} + 6t\mathbf{j}$ and $\mathbf{v} = \mathbf{i}$ at $t = 0$. Find $\mathbf{v}$ at time t .

$$\mathbf{v} = \int (2\mathbf{i} + 6t\mathbf{j}) dt = 2t\mathbf{i} + 3t^2\mathbf{j} + \mathbf{c}$$

At $t = 0$, $\mathbf{v} = \mathbf{i}$, so $\mathbf{c} = \mathbf{i}$ and

$$\mathbf{v} = (1 + 2t)\mathbf{i} + 3t^2\mathbf{j}$$

Speed is the magnitude of the velocity vector, so it is a scalar: $|\mathbf{v}| = \sqrt{v_1^2 + v_2^2}$. Quoting a "speed" as a vector loses the mark.

2. Distance and displacement in two dimensions

- **Distance from** the origin at time t is $|\mathbf{r}|$.
- **Displacement** between two times is the difference of the position vectors.
- **Distance travelled** is the length of the path, generally greater than the magnitude of the displacement.

Worked example. For $\mathbf{r} = 2t\mathbf{i} + t^2\mathbf{j}$, find the average velocity between $t = 0$ and $t = 2$.

Displacement $= (4\mathbf{i} + 4\mathbf{j}) - \mathbf{0} = 4\mathbf{i} + 4\mathbf{j}$, over 2 seconds:

$$\text{average velocity} = 2\mathbf{i} + 2\mathbf{j}$$

3. Constant acceleration in vector form

The suvat formulae carry over directly, with vectors in place of scalars (k10):

$$\mathbf{v} = \mathbf{u} + \mathbf{a}t$$

$$\mathbf{s} = \mathbf{u}t + \frac{1}{2}\mathbf{a}t^2$$

$$\mathbf{s} = \mathbf{v}t - \frac{1}{2}\mathbf{a}t^2$$

$$\mathbf{s} = \frac{1}{2}(\mathbf{u} + \mathbf{v})t$$

Worked example. A particle has $\mathbf{u} = 2\mathbf{i}$ and $\mathbf{a} = \mathbf{j}$. Find its displacement after 4 s.

$$\mathbf{s} = 2\mathbf{i}(4) + \frac{1}{2}\mathbf{j}(16) = 8\mathbf{i} + 8\mathbf{j}$$

Worked example. A particle starts at $2\mathbf{i} + \mathbf{j}$ with constant velocity $3\mathbf{i} - \mathbf{j}$. Find its position at $t = 4$.

$$\mathbf{r} = (2\mathbf{i} + \mathbf{j}) + 4(3\mathbf{i} - \mathbf{j}) = 14\mathbf{i} - 3\mathbf{j}$$

Because $\mathbf{v} = \mathbf{u} + \mathbf{a}t$, the **change** in velocity is always parallel to the acceleration. With zero acceleration the velocity is constant and the path is a straight line.

4. Relative position and relative velocity

position of B relative to $A = \mathbf{r}_B - \mathbf{r}_A$

velocity of A relative to $B = \mathbf{v}_A - \mathbf{v}_B$

Worked example. A is at $3\mathbf{i} + 4\mathbf{j}$ and B at $7\mathbf{i} + \mathbf{j}$.

Position of B relative to A is $4\mathbf{i} - 3\mathbf{j}$, and the distance between them is $\sqrt{16 + 9} = 5$.

Worked example. A has velocity $5\mathbf{i} + 2\mathbf{j}$ and B has $3\mathbf{i} - \mathbf{j}$. The velocity of A relative to B is

$$(5 - 3)\mathbf{i} + (2 - (-1))\mathbf{j} = 2\mathbf{i} + 3\mathbf{j}$$

Collisions

Two particles **collide** when their position vectors are equal *at the same instant* — that is, both components agree for a single common value of t .

Worked example. A has $\mathbf{r}_A = (t + 1)\mathbf{i} + 2t\mathbf{j}$ and B has $\mathbf{r}_B = (3t - 1)\mathbf{i} + (t + 1)\mathbf{j}$. Do they collide?

Equating $\mathbf{i}$ components: $t + 1 = 3t - 1$, so $t = 1$.

Equating $\mathbf{j}$ components: $2t = t + 1$, so $t = 1$.

Both give the same time, so they collide at $t = 1$, at the point $2\mathbf{i} + 2\mathbf{j}$.

You must check **both** components and confirm they give the *same* t . If the two components give different times, the paths cross but the particles are never there together — no collision.

The distance between two particles is least when the magnitude of the relative position vector is minimised, which is found by writing that magnitude as a function of t and minimising it.

5. The Cartesian equation of the path

Eliminate t between the component equations (k11).

Worked example. $\mathbf{r} = 3t\mathbf{i} + t^2\mathbf{j}$.

Here $x = 3t$ and $y = t^2$, so $t = \frac{x}{3}$ and

$$y = \frac{x^2}{9}$$

Worked example. $\mathbf{r} = 4t\mathbf{i} + (3t - 5t^2)\mathbf{j}$.

$x = 4t$ gives $t = \frac{x}{4}$, so

$$y = \frac{3x}{4} - \frac{5x^2}{16}$$

a parabola — the signature of motion under constant acceleration perpendicular to the initial velocity.

Worked example. For $\mathbf{r} = (t - 2)\mathbf{i} + (t^2 - 4)\mathbf{j}$, when does the particle pass through the origin? The $\mathbf{i}$ component vanishes at $t = 2$; the $\mathbf{j}$ component vanishes at $t = \pm 2$. Both are zero at $t = 2$, so it passes through the origin then.

6. Common pitfalls

- Reporting a speed as a vector rather than its magnitude.
- Omitting the vector constant of integration.
- Subtracting relative position or velocity the wrong way round.
- Concluding a collision from the $\mathbf{i}$ components alone.
- Confusing distance from the origin with distance travelled.
- Applying constant-acceleration formulae when the acceleration is a function of t .
- Eliminating t from the harder component and introducing a spurious square root.